

ICT 1201

Fundamentals of Computer
Systems

Introduction to
Programming and Programming
Languages

What is a computer Program?

- ❑ A computer program is a sequence of instructions written using a Computer programming Language to perform a specified task by the computer.
- ❑ These programs make the computer understand the user's requirements, the function to be performed and the output to be given to the user.

Uses of Computer Programs

- ❑ Today computer programs are being used in almost every field, household, agriculture, medical, entertainment, defense, communication, etc.
- ❑ MS Word, MS Excel, Adobe Photoshop, Internet Explorer, Chrome, etc., are examples of computer programs.
- ❑ Computer programs are being used to develop graphics and special effects in movie making.
- ❑ Computer programs are being used to perform Ultrasounds, X-Rays, and other medical examinations.
- ❑ Computer programs are being used in our mobile phones for SMS, Chat, and voice communication.

Why we need Programming Languages?

- ❑ In general you communicate with humans by using natural languages such as Sinhala, Tamil, English and so forth.
- ❑ If you want to communicate with a machine like a Computer, you can not use such languages.
- ❑ Language known as programming languages can be used to instruct computers.

Different types of programming languages

- ❑ Machine level language
- ❑ Assembly level language
- ❑ High level language

Machine Level Language

- ❑ Machine level language is the only language understood by computers.
- ❑ Program entirely consists of binary numbers (0's and 1's).
- ❑ Machine language instruction has two parts.

– E.g. 0000101 00001000



Operation code :
ADD

Such as addition
or subtraction



Operand :
8

That identify the
data to be
processed.

- ❑ Computer program written using any language should be translated to machine language before execute.

Assembly Level Language

- ❑ Symbolic operation codes replaced binary operation codes.
 - E.g. ADD A: Represent Addition
- ❑ Assemblers convert assembly programs to machine code.
- ❑ Easier to write.

High Level Language

- ❑ Instructions closer to everyday English.
- ❑ It is easy to understand and write the program.
- ❑ E.g. BASIC, COBOL, FORTRAN, C, Java, Visual Basic, etc.
- ❑ Following is a simple program written in C programming language.

```
/* Hello World program */  
  
#include<stdio.h>  
  
main()  
{  
    printf("Hello World");  
  
}
```


Home work

- ❑ Find out advantages and disadvantages of different types of programming languages.

Types of Programming Language	Advantages	Disadvantages
Machine Level Language		
Assembly Level Language		
High Level Language		

Language Processors / Translators

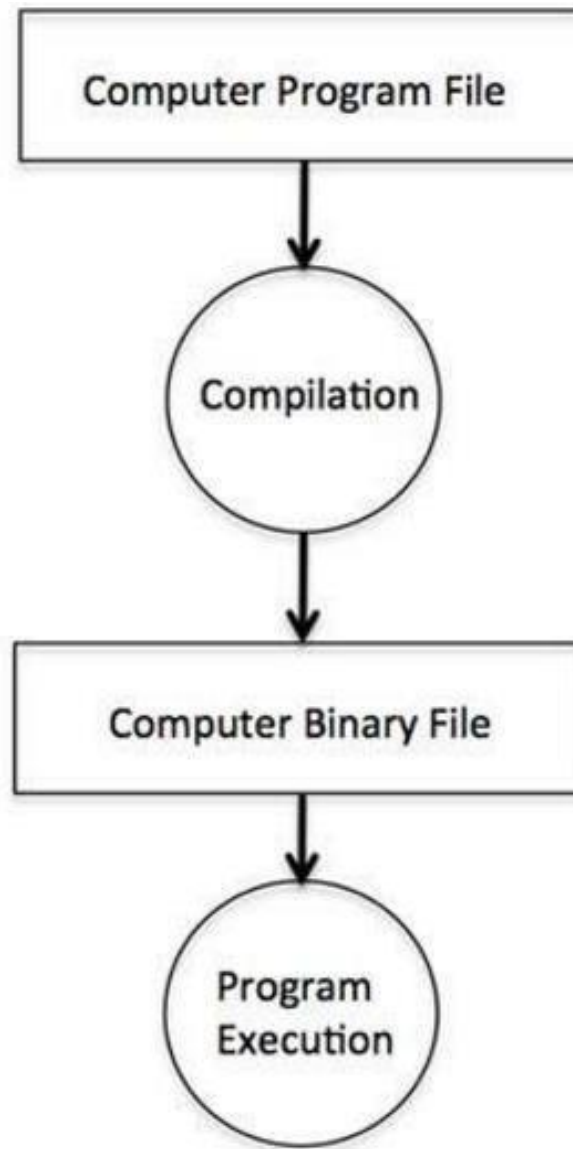
- ❑ Computers can understand only one language.
- ❑ It is known as machine language.
- ❑ However humans are not familiar with the machine language.
- ❑ So you have to specify your idea using a high level programming language first.
- ❑ Then translate it into machine language.

Language Processors / Translators

- ❑ In order to convert the High level Language to Machine Level language, Language Processors are used.
- ❑ Language processors are two types.
 - Compilers
 - Interpreters

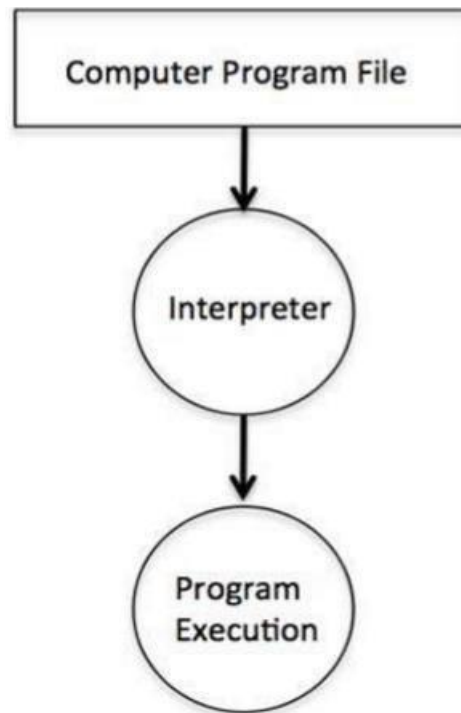
Compiler

- ❑ Compiler translates the entire program code from the language it is written in, into machine level language.
- ❑ This process of conversion from text formatted program to binary format file is called program compilation.
- ❑ If you are going to write your program in any such language, which needs compilation like C, C++, Java and Pascal, etc., then you will need to install their compilers before you start programming.



Interpreter

- ❑ An interpreter translates program code to machine level language on line by line basis.

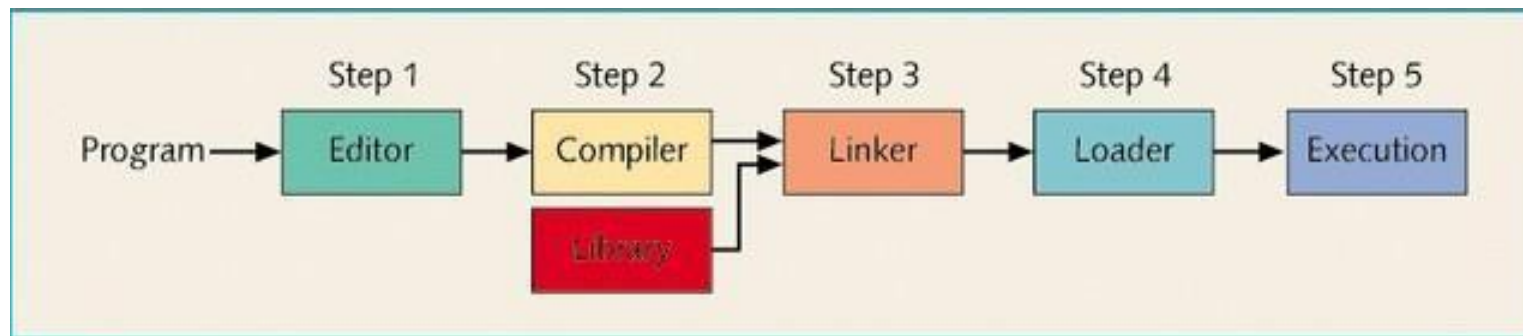


Difference between an interpreter and compiler

Interpreter	Compiler
Translate program one statement at a time.	Scans the entire program and translate it as a whole into machine code.
It takes less amount of time to analyze the source code but the overall execution time is slower.	It takes large amount of time to analyze the source code but overall execution time is comparatively faster.
No intermediate object code(Binary File) is generated.	Generates intermediate object code (Binary file).
Continues translating the program until the first error is met, in which case it stops. Hence debugging is easy.	It generates the error message only after scanning whole program.
Programming language like Python, Ruby, PHP use this.	Programming language like C,C++ use Compilers.

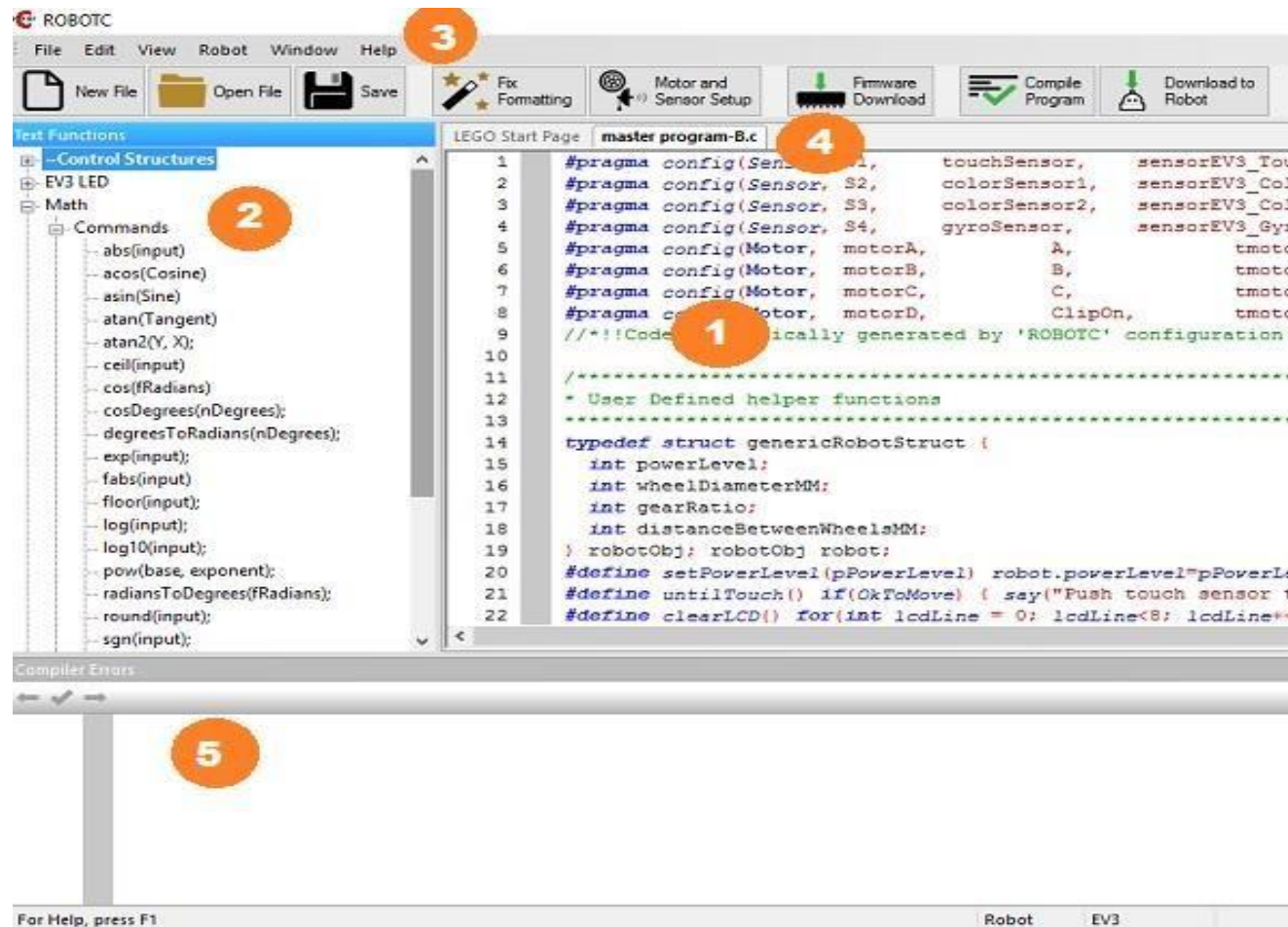
Processing a High-Level Language Program

- ❑ Source program created with an editor
- ❑ Source code translated into machine language by compiler
 - ❑ Results in a .obj file (object code)
- ❑ Linker combines common library routines with object code
 - ❑ Results in a .exe file (executable code)
- ❑ Loader brings executable code into memory and it is run



Integrated Development Environment (IDE)

- IDE is a software application that provides comprehensive facilities to computer programmers for software development.
- Typically, an IDE contains a code editor, a compiler or interpreter and a debugger that the developer accesses through a single graphical user interface.



1. Editor
2. CodeTemplates
3. Toolbar
4. Programs Tab
5. ErrorMessage

Example IDEs

Programming Language	IDE
C and C++	Code::Blocks, Dev-C++, Qdeveloper, MonoDevelop
Java	NetBeans, BlueJ, Eclipse, Greenfoot, Jdeveloper, Xcode, Visual J++
PHP	Aptana, NetBeans, Kdevelop, PhpStorm
Python	Eric, Wing IDE, Light Table, Kdevelop, Ninja-IDE
Pascal	Delphi IDE, Lazarus

Components of a programming language

- ❑ English has a predefined grammar, which needs to be followed to write English statements in a correct way.
- ❑ Likewise, most of the Human Languages (Tami, Hindi, English, etc.) are made of several elements like verbs, nouns, adjectives, adverbs, propositions, and conjunctions, etc.
- ❑ Similar to that Computer Programming Languages are also made of basic three elements.

Components of a programming language

- ❑ Vocabulary
 - ❑ A collection of terms valid for the language.
 - ❑ It consists of preserved words provided by the language and user defined terms.
- ❑ Syntax
 - ❑ Defines the rules and guidelines for constructing sentences or instructions.
- ❑ Semantics
 - ❑ The meaning of the language instructions.

Steps to Develop a Program

- ❑ Developing a program involves steps similar to any problem solving task.
- ❑ There are five main ingredients in the programming process.
 - Defining the problem
 - Planning the solution
 - Coding the problem
 - Testing the program
 - Documenting the program

Defining the Problem

- ❑ Defining the problem consists of identifying what is known and what is wanted to obtain.
- ❑ Eventually, the programmer produces a written agreement that, among other things, specifies the kind of input, processing and output required.

Planning the solution

- ❑ This is the stage where you decide how your program will work to meet the decisions you made during defining the problem.
- ❑ In this phase you simply write down the steps in the correct order, that you are required to perform, to solve the problem.
- ❑ When all these steps are put together you have what is called an Algorithm.

What is an Algorithm?

- ❑ An algorithm is a complete step-by-step procedure for solving a problem or accomplishing a task.
- ❑ Algorithms can be represent by using
 - ❑ Flow charts.
 - ❑ Pseudo Codes.

What is an Algorithm?

- ❑ The following is an algorithm for making a cup of tea.
 1. Fill the kettle with water.
 2. Boil the water in the kettle.
 3. Put the tea bag in a cup.
 4. Pour some of the boiled water into the cup.
 5. Add milk to the cup.
 6. Add sugar to the cup.
 7. Stir the tea.

Exercise

- ⑩ Write algorithms for following problems.
 - ✎ Find the sum of two numbers and display it.
 - ✎ Find the area of triangle and display it.
 - ✎ Find the largest from given two numbers and display it.

Answers

- ❑ Find the sum of two numbers and display it.
 1. Input two numbers.
 2. Take the summation.
 3. Print the summation.

Answers

- ❑ Find the area of triangle and display it.
 1. Input values for base and height.
 2. Take the area by multiplying $0.5 * \text{base} * \text{height}$.
 3. Display the area

Answers

- ❑ Find the largest from given two numbers and display the largest number.
 1. Input two numbers.
 2. Check the largest number.
 3. Print the largest number.

Flowcharts

- It is a step by step diagrammatic representation of the program.
- Each type of task is represented by a symbol.

Diagram	Notation	Representation
	Oval	Start/ End of the Program
	Parallelogram	Input/output of Data
	Rectangle	Processing Operation
	Rhombus	Decision Box

Flowcharts

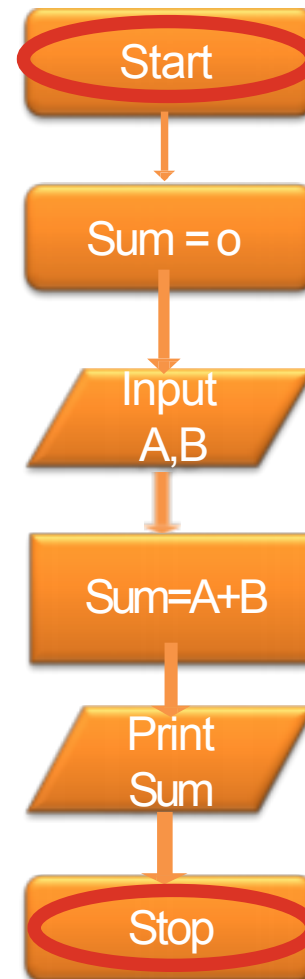
Diagram	Notation	Representation
	Circle	Connection
	Flow Lines	Direction of Flow

Flowcharts - Example

- Find the sum of two numbers and display it.

Algorithm

1. Input two numbers.
2. Take the summation.
3. Print the summation



Flowcharts

- ❑ Draw flowcharts for followings.
 - ❑ Find the area of triangle and display it.
 - ❑ Find the largest from given two numbers and display it.

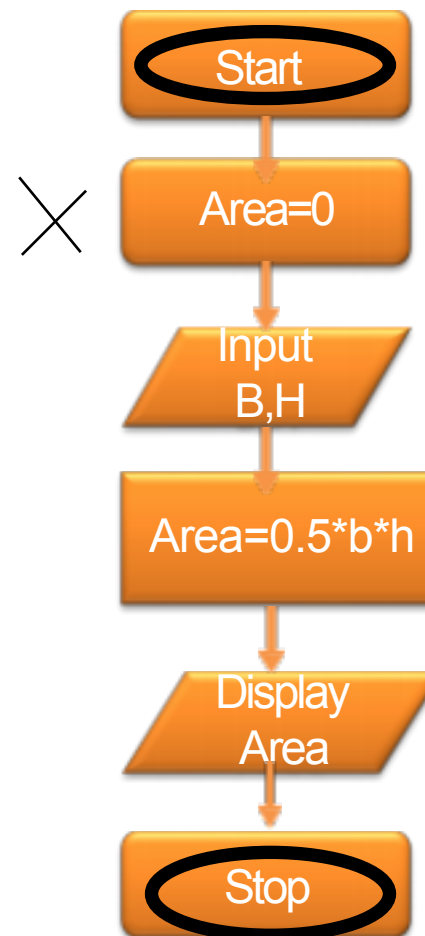
Answers

- Find the area of triangle and display it.

Algorithm

1. Input values for base and height.
2. Take the area by multiplying $0.5 * \text{base} * \text{height}$.
3. Display the area

Flowchart



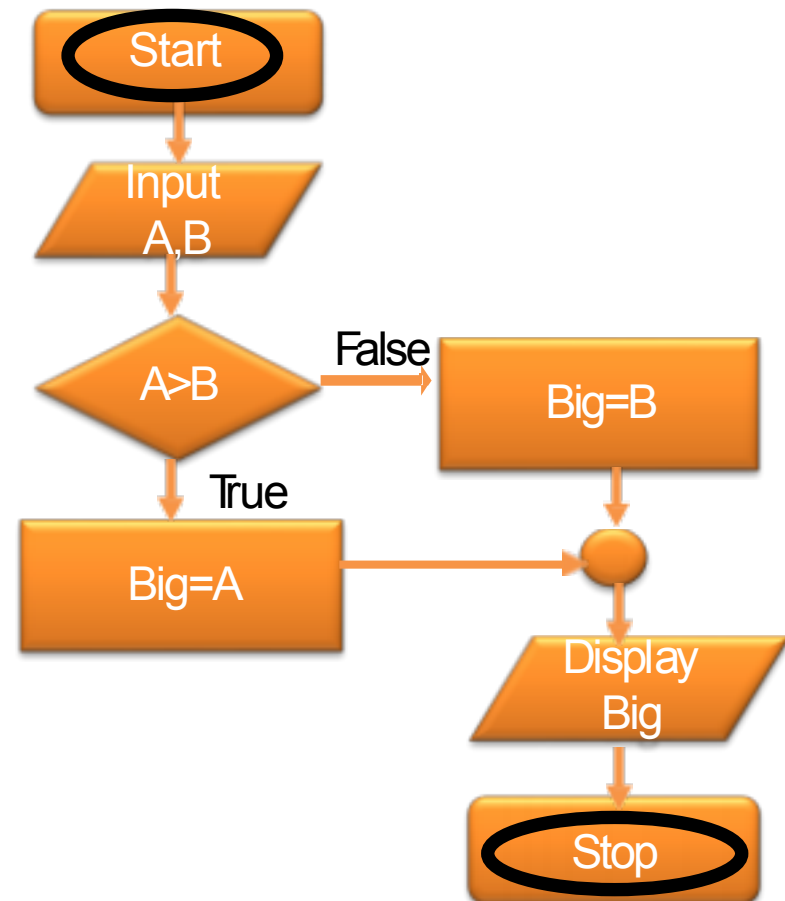
Answers

- Find the largest from given two numbers and display the largest number.

Algorithm

1. Input two numbers
2. Check the largest.
3. Print the largest number.

Flowchart



Pseudo code

- Pseudo code is the outline of a program, written in a form that can be easily converted into high level program statement.

Pseudo code - Example

- Find the sum of two numbers and display it.

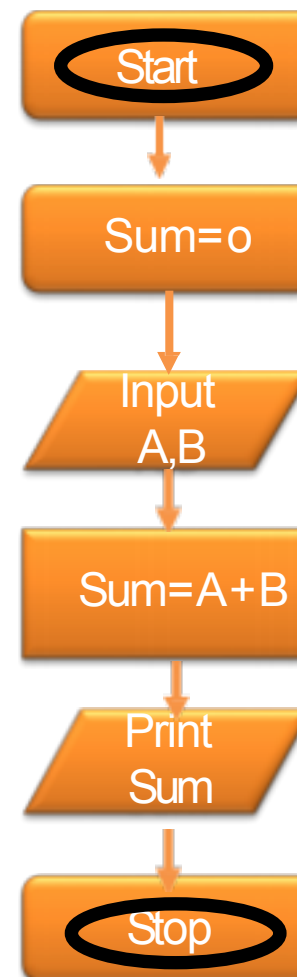
Algorithm

- 1.Input two numbers.
- 2.Take the summation.
- 3.Print the summation.

Pseudo code

- 1.INPUT A
- 2.INPUT B
- 3.Sum = A+B
- 4.OUTPUT Sum

Flowchart



- ❑ As the programmer, the next step is to code the program that is, to express a solution in a programming language.
- ❑ The programmer will translate the logic from the flowchart or pseudo code or some other tool to a programming language.
- ❑ To get the program to work, the programmer has to follow exactly the rules, the syntax – of the language used.

Testing the program

- ❑ After coding the program, you must test it on the computer.
- ❑ This step involve these phases:
 - ❑ Desk – checking : simply sit down and mentally trace, or check, the logic of the program to attempt to ensure that the code is error-free and workable. (in industry this is known as peer - review)

Testing the program

- ✓ Translating – a translator is a program that: Checks the syntax of the program to make sure the programming language was used correctly, giving all the syntax – error messages, called diagnostics, and then Translates the program into form the computer can understand. The mistakes are called syntax errors, The translator provides the descriptive error messages. The translation involves the original program, called a source module, which is transformed by a computer into an object module.
- ✓ Debugging – debugging means detecting, locating and correcting bugs (mistakes), usually by running the program. These bugs are logic errors.

Documenting the program

- ❑ Documentation is a detailed written description of the programming cycle and specific facts about the program.
- ❑ Comments in the program itself are also considered an essential part of documentation.

Exercise

- Write down the algorithms and draw flowcharts for given scenarios.
- 1. A company gives discount for the total bill paid by the customers as follows.
 - If the bill amount is above Rs.5000 a discount of 10% is given.
 - Otherwise 5% is given.
- Input the bill amount and calculate the discount amount.

Exercise

- Write down the algorithms and draw flowcharts for given problems.

2. Enter marks of four subjects and find the average. Display the grading as follows.

- If average is ≥ 90 , A
- If average is ≥ 80 , B
- If average is ≥ 70 , C
- If average is ≥ 50 , D
- If average is < 50 , F

THANK YOU